

TURGAY, Kazakhstan

WMO: 353580

Lat: 49.6347N

Lon: 63.4822E

Elev: 436

StdP: 14.47

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.6	-17.1	-28.7	1.1	-21.1	-23.8	1.5	-16.9	26.9	13.3	20.9	3.7	6.1	20	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	96.7	67.9	93.4	66.9	90.1	65.8	69.3	92.8	67.8	90.4	66.4	88.3	5.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.3	79.4	80.6	58.2	73.5	80.0	56.2	68.3	79.2	33.6	92.9	32.3	90.7	31.2	88.7	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	17.1	15.8	-24.1	102.9	4.7	3.3	-27.5	105.3	-30.3	107.2	-33.0	109.0	-36.4	111.4
			-24.4	72.4	4.7	2.4	-27.8	74.1	-30.5	75.5	-33.1	76.9	-36.5	78.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.8	3.8	5.0	23.6	48.4	63.0	72.4	75.1	73.3	61.4	45.4	27.9	11.5
	DBStd	27.96	12.69	12.26	12.66	10.51	7.91	7.40	6.31	7.57	8.67	8.89	11.61	12.86
	HDD50	5734	1432	1259	818	153	5	0	0	0	10	198	664	1195
	HDD65	9087	1897	1679	1282	501	134	20	5	19	166	610	1114	1660
	CDD50	3096	0	0	1	105	408	672	779	723	354	54	0	0
	CDD65	975	0	0	0	2	72	242	320	278	60	1	0	0
	CDH74	11286	0	0	0	46	831	2868	3733	3159	634	15	0	0
	CDH80	5095	0	0	0	4	264	1319	1775	1501	230	2	0	0
Wind	WSAvg	6.0	6.0	6.5	6.3	6.2	6.9	6.4	6.3	5.7	5.5	5.5	5.1	5.6
Precipitation	PrecAvg	8.2	0.5	0.4	0.5	0.7	0.6	0.7	0.7	0.7	0.6	0.9	0.8	0.8
	PrecMax	12.3	1.5	1.2	1.8	2.3	1.4	4.3	2.9	2.4	1.7	3.1	2.2	1.9
	PrecMin	3.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.9	0.4	0.3	0.4	0.7	0.4	0.9	0.7	0.7	0.5	0.7	0.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.8	33.5	61.3	79.2	90.8	99.7	101.2	100.6	93.6	75.5	53.1	35.8
		MCWB	30.5	30.1	47.8	60.2	65.2	68.4	69.1	68.0	66.4	56.9	45.4	31.2
	2%	DB	30.8	30.7	50.5	73.5	86.4	95.3	96.5	95.9	86.3	70.0	48.2	33.3
		MCWB	28.3	27.5	42.1	57.3	63.6	66.9	68.0	68.0	62.9	53.7	42.3	30.4
	5%	DB	27.9	27.4	43.3	69.3	82.6	91.4	92.5	92.2	80.9	64.9	44.0	31.4
		MCWB	25.5	24.6	37.8	55.1	62.0	66.0	66.8	66.7	60.9	51.3	39.1	28.7
	10%	DB	23.4	23.4	38.4	65.0	78.4	87.5	89.1	88.4	76.4	60.1	40.5	28.9
		MCWB	21.2	21.1	33.9	53.2	60.2	64.5	65.4	65.2	58.7	48.8	36.4	26.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.0	30.4	49.1	61.6	66.6	71.4	72.1	70.8	67.4	58.2	47.3	32.6
		MCDB	32.6	33.4	60.0	76.6	86.1	95.9	95.4	94.7	90.8	74.7	52.1	34.5
	2%	WB	28.3	27.6	43.0	58.5	64.4	68.3	69.3	69.0	63.9	54.5	42.6	30.7
		MCDB	30.6	30.5	50.2	71.9	85.2	91.5	92.4	92.5	85.0	68.2	47.4	33.1
	5%	WB	25.5	24.8	38.0	56.1	62.6	66.6	67.6	67.3	61.2	51.9	39.4	28.6
		MCDB	27.8	27.4	43.2	67.8	80.7	89.3	89.8	90.1	79.7	63.9	43.4	31.3
	10%	WB	21.4	21.2	34.1	53.3	60.6	65.0	66.2	65.8	58.9	49.1	36.4	26.2
		MCDB	23.4	23.4	38.2	64.5	77.6	86.4	87.7	87.4	75.7	59.7	40.5	28.8
Mean Daily Temperature Range	5% DB	MDBR	13.3	14.6	15.4	19.5	22.2	22.5	22.1	22.1	21.8	19.0	11.9	12.7
		MCDBR	11.8	15.0	20.2	25.7	26.3	26.9	26.6	26.7	26.8	27.2	15.5	13.3
		MCWBR	11.0	13.6	14.7	15.1	13.2	11.4	11.2	11.6	13.0	16.8	11.5	11.1
	5% WB	MCDBR	12.2	14.6	19.8	24.5	25.0	25.6	24.7	25.6	25.9	25.6	14.4	12.0
		MCWBR	11.5	13.3	14.7	15.6	13.3	12.1	11.5	12.1	13.6	16.4	11.3	10.7
Clear-Sky Solar Irradiance	taub	0.307	0.333	0.397	0.424	0.420	0.417	0.415	0.412	0.363	0.351	0.340	0.312	
	taud	2.078	2.018	2.014	2.087	2.178	2.246	2.268	2.254	2.342	2.291	2.191	2.112	
	Ebn at Noon	221	246	250	259	266	268	266	260	263	242	209	200	
	Edh at Noon	27	37	45	47	44	42	40	39	32	28	25	23	
All-Sky Solar Radiation	RadAvg	471	851	1242	1578	2045	2170	2044	1795	1367	801	390	308	
	RadStd	66	87	139	137	116	118	114	126	80	82	44	45	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page